

LoFi Granulator – User Guide

The LoFi Granulator is an instrument that takes an audio sample from an SD card and cuts it into small fragments for playback. These fragments can be played in order or out of order, forwards or backwards, tuned higher or lower, and muted or unmuted.

Getting started:

- Ensure that the SD card with WAV files is inserted.
- Power the Granulator via USB or batteries and turn on.
- It may take a few seconds for the sounds to load from the SD card.
- Press the **Green button** to start playback. Press again to stop.
- Playback loops around the sample such that sound always heard.

Changing sounds:

The Granulator selects the first (alphabetical) WAV file on the SD card at startup. To select the next file, **hold the Blue button and press the Green button**. Loading files can take a few seconds, during loading there will be no sound. Repeat this to cycle through successive files; the selection will loop back to the first file at the end of the list.

Randomising grain parameters:

An easy way to explore the possibilities of the LoFi Granulator is to **click the Blue button** to randomise the parameters. **Long Hold the Blue button** to reset parameters to startup state.

Changing the grain parameters:

There are 4 dials on the LoFi Granulator. These adjust 8 grain parameters, four directly and another 4 when the Blue button is held down as a 'shift' key. Note that dials 'latch' to the previous value when the blue (shift) key is pressed and released, so sweeping a dial will make sure it is active.

Dials are numbered 1-4 from the bottom left corner, counting clockwise.

- Dial 1 – Grain Size / Grain Start Position diversity
- Dial 2 – Grain Pitch / Grain Pitch diversity
- Dial 3 – Playback Position within the sample / Playback speed and direction
- Dial 4 – Grain Density / Volume

Adding sounds to the SD card:

The LoFi Granulator will play any valid WAV file up to its memory limit. This is approximately 10 seconds of mono audio at 44.1k sample rate, 16bit resolution. If longer files are used, the Granulator will load as much of it as possible from the start and truncate the rest.

Many audio sample rates and bit depths are supported, but 16bit, 44.1k is optimal.

Load files onto the SD card from your computer. The SD card is formatted as FAT32, so should be readable on a PC, Mac or Linux computer.

Experiment with different sound sources to explore the variety of sonic possibilities of the Granulator.

Effects:

The LoFi Granulator includes some effects, delay and reverberation.

- Subtle reverberation is always on.
- Delay amount increases as the density decreases.

Problem solving:

If there is no sound, then;

- Power down and restart the Granulator.
- Check the SD card is inserted and has WAV files on it.
- Ensure the volume is turned up.
- See if any internal wire connections have dislodged.

Updating:

Documents about the LoFi Granulator are available on GitHub, including this user guide, the assembly instructions, and latest software update:

<https://github.com/algomusic/lofi-granulator>

Some tips for getting started:

When the Granulator is first starts, parameters are set so that grains play in order and read through the sample at the 'normal' playback rate. So you should hear the sample in its original form. From that starting point, here are some pathways to explore.

Break beat cutup: With a simple drumbeat sample playing at startup, adjust the Grain Start Position Diversity (shift-dial1) to rearrange the order of nearby gains. Return the dial to minimum to restore the grain order. Change the grain size (dial1) to experiment with different fragment subdivisions.

Freezing a sample position: With a vocal sample playing, to make the effect easy to hear, adjust the playback rate (shift-dial3) to the middle. In the centre, grains will be chosen from a stationary location in the sample. Adjusting the dial to the right will increase the playback rate forward through the sample and adjusting it to the left of centre will move playback backwards through the sample and play gains in reverse. With the rate frozen, adjust the playback position (dial3) within the file. Experiment with the grain size (dial1) to get different textures.

Independent pitch and speed: One of the features of granular synthesis is being able to decouple playback speed from playback pitch. The freezing tip above modified playback speed. To adjust the pitch of playback, move dial2. In the centre, playback will be 'normal', adjusting to the right will increase the pitch of each grain and to the left of centre pitch will get lower. Use different combinations of playback rate (shift-dial3) and pitch (dial2) to explore these decoupled sonic features. Try increasing and decreasing the gain density (dial4) for more variety.